

1 . A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

Given:

- Box A: 6 Red, 4 Blue, 2 Green $\rightarrow$ Total = 12
- Box B: 5 Red, 7 Blue, 3 Green $\rightarrow$ Total = 15
- Box C: 8 Red, 2 Blue, 5 Green $\rightarrow$ Total = 15

Since Box B is twice as likely to be selected:

- $P(A) = \frac{1}{4}$
- $P(B) = \frac{1}{2}$
- $P(C) = \frac{1}{4}$

Probability of drawing a blue ball:

- $P(\text{Blue}|A) = \frac{4}{12} = \frac{1}{3}$
- $P(\text{Blue}|B) = \frac{7}{15}$
- $P(\text{Blue}|C) = \frac{2}{15}$

Using Bayes' theorem,

$$P(C|\text{Blue}) = \frac{P(\text{Blue}|C)P(C)}{P(\text{Blue}|A)P(A)+P(\text{Blue}|B)P(B)+P(\text{Blue}|C)P(C)}$$

Substitute values:

$$= \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{1}{3} \times \frac{1}{4} + \frac{7}{15} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4}} = \frac{\frac{1}{30}}{\frac{1}{12} + \frac{7}{30} + \frac{1}{30}} = \frac{\frac{1}{30}}{\frac{7}{20}} = \frac{2}{21}$$

$$P(\text{Box C}|\text{Blue}) = \frac{2}{21} \approx 0.0952$$

Probability = 9.52%

2. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data.

Calculate: Accuracy Precision Sensitivity Specificity.

Given:

- TP = 180

- $TN = 160$
- Each class = 1200 samples
- Test data = 25%

Step 1: Test Samples

Total dataset

$$1200 + 1200 = 2400$$

Test samples

$$25\% \times 2400 = 600$$

Actual Positive = 300

Actual Negative = 300

Step 2: Find FN and FP

$$FN = 300 - 180 = 120 \quad FP = 300 - 160 = 140$$

Confusion Matrix:

Predicted Positive Predicted Negative

Actual Positive $TP = 180$

$FN = 120$

Actual Negative $FP = 140$

$TN = 160$

Accuracy

$$\frac{TP+TN}{TP+TN+FP+FN} = \frac{180+160}{600} = \frac{340}{600} = 0.5667$$

Accuracy = 56.67%

Precision

$$\frac{TP}{TP+FP} = \frac{180}{180+140} = \frac{180}{320} = 0.5625$$

Precision = 56.25%

Sensitivity (Recall)

$$\frac{TP}{TP+FN} = \frac{180}{180+120} = \frac{180}{300} = 0.60$$

Sensitivity = 60%

Specificity

$$\frac{TN}{TN+FP} = \frac{160}{160+140} = \frac{160}{300} = 0.5333$$

Specificity = 53.33%

Final Answers

- **Accuracy = 56.67%**
- **Precision = 56.25%**
- **Sensitivity = 60.00%**
- **Specificity = 53.33%**

3. A deep learning model is trained for a classification task, and the following loss values are observed:

- Epoch 1: Training Loss = 2.8, Validation Loss = 2.6
- Epoch 2: Training Loss = 2.4, Validation Loss = 2.2
- Epoch 3: Training Loss = 2.1, Validation Loss = 1.9
- Epoch 4: Training Loss = 1.9, Validation Loss = 1.8
- Epoch 5: Training Loss = 1.7, Validation Loss = 1.9
- Epoch 6: Training Loss = 1.5, Validation Loss = 2.1
- Epoch 7: Training Loss = 1.3, Validation Loss = 2.4
- Epoch 8: Training Loss = 1.2, Validation Loss = 2.7

Answer the following:

- At which epoch does the model achieve the best generalization performance?
- At which point does overfitting begin? Justify your answer.
- Suggest two techniques to reduce overfitting and briefly explain how they help.

Epoch	Training Loss	Validation Loss
1	2.8	2.6
2	2.4	2.2
3	2.1	1.9
4	1.9	1.8
5	1.7	1.9
6	1.5	2.1
7	1.3	2.4
8	1.2	2.7

(a) Best Generalization Performance

The model performs best when the **validation loss is minimum**.

Minimum validation loss = **1.8** at **Epoch 4**.

Answer:

Epoch 4 achieves the best generalization performance.

(b) When does overfitting begin?

Overfitting starts when:

- Training loss continues to decrease.
- Validation loss begins to increase.

After **Epoch 4**:

- Training loss decreases ($1.9 \rightarrow 1.7$).
- Validation loss increases ($1.8 \rightarrow 1.9$).

Therefore, **overfitting begins from Epoch 5**.

Answer:

Overfitting begins at **Epoch 5** because the training loss keeps decreasing while the validation loss starts increasing.

(c) Two techniques to reduce overfitting

1. Dropout

- Randomly deactivates neurons during training.
- Prevents the network from relying too much on specific neurons, improving generalization.

2. Early Stopping

- Stops training when the validation loss starts increasing.
- Prevents the model from learning noise in the training data and reduces overfitting.